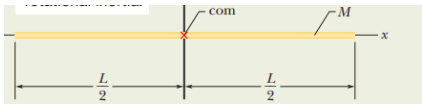
BRAC University

Department of Mathematics & Natural Sciences

Principles of Physics-I (PHY 111)

Assignment-III Summer, 2024



1. In figure shows a thin, uniform metal rod of mass M and length L, on an x axis with the origin at the rod's center.

   (a) What is the rotational inertia of the rod about the perpendicular rotation axis through the center?

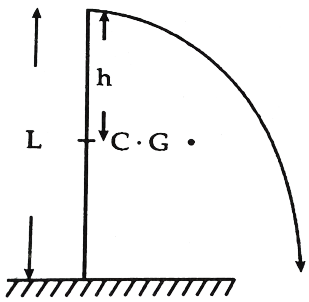
   (b) What is the rod's rotational inertia I about a new rotation axis that is perpendicular to the rod and through the left end?

   (c) Now consider the rod's cross-sectional area is $0.001m^2$ and mass $M = 1kg$. If the rod is melted and formed into a sphere, what is the magnitude of the rotational inertia (consider here the rotational axis is the diameter of the sphere)? **(3+2)**

   [3+2]

2. (a) Find the moment of inertia I of a uniform coin of mass M with radius R rotating about the perpendicular line through the center.

   (b) What is the moment of inertia if you strike this coin on the table when it is rotating about its diameter? [Hint: Use perpendicular axis theorem.] **(3+2)**



3. A rod of uniform mass M and length L is positioned vertically on the ground with one end touching the ground. When it falls, it rotates without slipping.

   (a) By using the formula $I = \int r^2 dm$, find the moment of inertia of the rod in the above case.

   Now consider the rod's mass $M = 2.0kg$, length $L = 1m$, and the gravitational acceleration $g = 9.8m/s^2$.

   (b) What is the magnitude of the angular velocity (ω) of the rod while it is striking at the ground?

   (c) What is the magnitude of the velocity (v) of the upper tip when it hits the ground? **(2+2+1)**